Section 15.6: Triple Integrals

Multiple Integrals

From Double to Triple

In Sections 15.1–15.5 we integrated $f(x, y)$ over a 2D region in $\mathbb{R}^2$. Now we go one dimension higher: integrating $f(x, y, z)$ over a 3D solid in $\mathbb{R}^3$.

Recall the progression:

- **Calc I:** $\displaystyle\int_a^b f(x)\,dx$ — integrate over an interval (1D)
- **Calc III (Ch 15.1–15.5):** $\displaystyle\iint_D f(x,y)\,dA$ — integrate over a region (2D)
- **Now:** $\displaystyle\iiint_E f(x,y,z)\,dV$ — integrate over a solid (3D)

The construction is the same: partition the solid E into tiny boxes, multiply each f-value by the volume of the box $\Delta V = \Delta x\,\Delta y\,\Delta z$, add them all up, and take the limit:

$$\iiint_E f(x,y,z)\,dV = \lim_{l,m,n\to\infty} \sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*)\,\Delta V$$

Same idea, one more dimension. Chop, multiply, add, take the limit.

Interpretation: Density and Mass

What does $\iiint_E f(x,y,z)\,dV$ *mean* physically?

If $f(x,y,z) = 1$ for all points, then:

$$\iiint_E 1\,dV = \text{volume of } E$$

This is the 3D version of “area $= \iint_D 1\,dA$.”

The most common physical interpretation: if $f(x,y,z) = \rho(x,y,z)$ is a **density function** (mass per unit volume, like $\mathrm{g/cm^3}$ or $\mathrm{kg/m^3}$), then

$$\iiint_E \rho(x,y,z)\,dV = \text{total mass of } E$$

More generally, if f gives “amount per unit volume,” then $\iiint_E f\,dV$ gives the total amount in E:

- Charge density $(\mathrm{C/m^3}) \to$ total charge
- Bacteria concentration $(\mathrm{cells/cm^3}) \to$ total bacteria count
- Probability density $\to$ total probability

Triple integrals accumulate a quantity spread throughout a 3D solid — density times volume gives mass.

Quick Check: Interpret a Triple Integral

Suppose $\rho(x,y,z) = 3z$ gives the density (in g/cm^3) of a solid E.

What does $\displaystyle\iiint_E 3z\,dV$ *compute? What are the units?*

Think about what density $\times$ volume gives you. What role does z play in the density?

Setting Up the Limits of Integration

Just like double integrals, we evaluate triple integrals as **iterated integrals**. There are **six possible orders** (we'll discuss this more shortly), but the strategy is the same for all of them.

The key idea: Choose one variable to integrate first (the **inner** integral). Its limits may depend on the other two variables. Then set up a double integral over the remaining two variables — exactly like Sections 15.2–15.3.

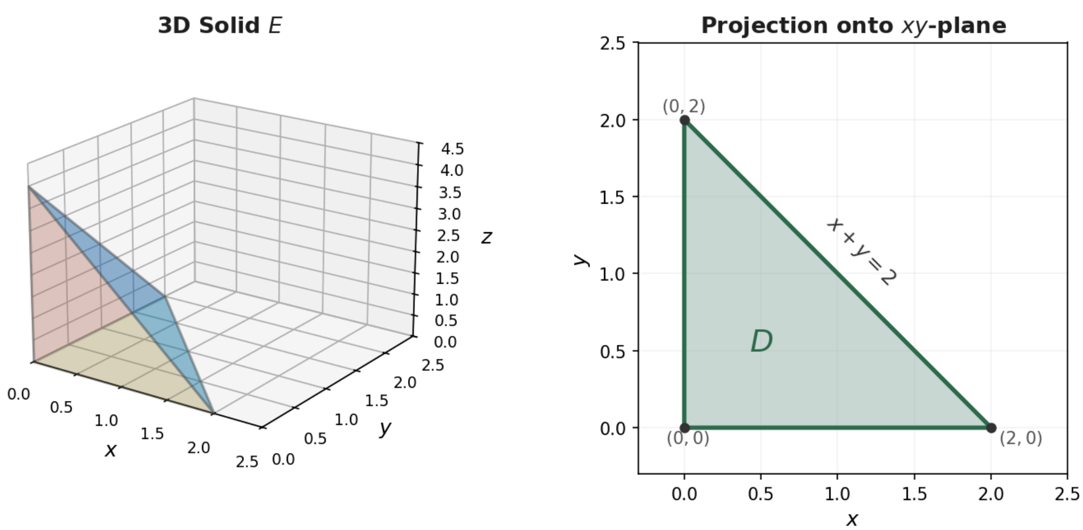
A common approach (inside out):

1. Pick a direction to “slice” the solid. For each fixed point in the other two variables, find where the solid starts and stops in the slicing direction $\to$ limits on the **inner** integral.
2. **Project** the solid onto the plane of the other two variables $\to$ a 2D region.
3. Set up the double integral over that 2D region as in Sections 15.2–15.3.

For example, if we slice in the z-direction:

$$\iiint_E f\,dV = \iint_D \left[\int_{z_{\text{bottom}}}^{z_{\text{top}}} f\,dz\right] dA$$

where D is the **shadow** of E on the xy-plane.



The hardest part of triple integrals is **finding the limits**. The actual integration is just three rounds of Calc I antiderivatives.

Example 1: Triple Integral Over a Tetrahedron

Example 1

Evaluate $\iiint_E 2xyz\,dV$, where E is the solid in the first octant bounded by the plane $3x + 2y + z = 1$.

Find the limits for $dz\,dy\,dx$. Start with: for fixed x, y, z goes from 0 up to the plane. Then project onto the xy-plane.

Quick Check: Describe the Solid

Describe the solid E whose volume is given by:

$$\int_0^1 \int_0^{1-x} \int_0^{2-2x-2y} dz\,dy\,dx$$

Read the limits from outside in. What shape has these boundary surfaces?

Six Orders of Integration

A triple integral can be written in **six** different orders:

$$dz\,dy\,dx \qquad dz\,dx\,dy \qquad dy\,dz\,dx \qquad dy\,dx\,dz \qquad dx\,dz\,dy \qquad dx\,dy\,dz$$

All six give the same answer (assuming the function is continuous), but some orders may be **much easier** to evaluate than others!

Strategy for choosing the best order:

- Look for limits that are **constants** — those variables should go on the outside

- Look for an integrand that simplifies when you integrate a particular variable first
- Sketch the solid and look for natural “top/bottom,” “left/right,” or “front/back” descriptions

For the tetrahedron below $3x + 2y + z = 1$ (Example 1), here's the $dy\,dz\,dx$ version:

Fix x and z: y goes from 0 to $\frac{1-3x-z}{2}$. **Fix x:** z goes from 0 to $1 - 3x$. **Then:** x goes from 0 to $\frac{1}{3}$.

$$\int_0^{1/3} \int_0^{1-3x} \int_0^{(1-3x-z)/2} 2xyz\,dy\,dz\,dx$$

Switching the order of integration changes the **limits**, not the **integrand**. Always re-derive the limits from scratch by thinking about the geometry.

Example 2: Volume Using a Triple Integral

Example 2

Find the volume of the solid bounded by the parabolic cylinder $y = x^2$, and the planes $z = 0$, $z = 4$, and $y = 9$.

Set up $\iiint_E 1\,dV$ in order $dz\,dy\,dx$. What are the top/bottom surfaces? Where does $y = x^2$ intersect $y = 9$?

Example 3: Setting Up in Two Different Orders

Example 3

The solid E is bounded by $z = 0$, $y = 0$, $z = y$, and $y = 4 - x^2$.

(a) Set up $\iiint_E 1\, dV$ in the order $dz\, dy\, dx$.

(b) Set up the same integral in the order $dy\, dz\, dx$.

For (a): What are the top and bottom surfaces? Project onto the xy-plane. For (b): What are the left and right surfaces in the y-direction? Project onto the xz-plane.

Homework

Section 15.6: 1, 3, 5, 6, 7, 9, 11, 13, 15, 17, 19, 25, 31, 33